

Phase 2 - SQL Queries, Output & Analytical Logic

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1. Question E3: Average Engagement by Platform

Logic: Grouped the dataset by 'platform' and calculated the mean for likes, shares, and comments using the AVG() aggregate function. A derived column summed these three metrics to calculate overall engagement, sorting the output in descending order.

SQL Query:

```
SELECT platform, AVG(likes) AS avg_likes, AVG(shares) AS avg_shares,
AVG(comments) AS avg_comments, AVG(likes + shares + comments) AS avg_total_engagement
FROM social_data GROUP BY platform ORDER BY avg_total_engagement DESC;
```

Output Preview:

platform	avg_likes	avg_shares	avg_comments	avg_total_engagement
YouTube	2312.04	1008.13	505.66	3825.84
Facebook	2330.93	984.46	506.70	3822.10
Reddit	2280.77	1002.69	509.47	3792.95
Instagram	2233.21	1039.41	498.72	3771.35
Unknown	2235.71	995.78	495.15	3726.65
Twitter	2209.83	1004.78	507.45	3722.07

2. Question M2: Do High Follower Users Get More Engagement?

Logic: Utilized a CASE statement to segment users into two distinct cohorts: 'High Follower' (>= 25,000) and 'Low Follower' (< 25,000). Calculated the distinct user count, total posts, and the AVG() of total engagement per group.

SQL Query:

```
SELECT CASE WHEN follower_count >= 25000 THEN 'High Follower (>= 25k)' ELSE 'Low Follower (< 25k)' END AS
user_category,
COUNT(DISTINCT user_id) as num_users, COUNT(post_id) as total_posts, AVG(likes + shares + comments) AS
avg_engagement_per_post
FROM social_data GROUP BY user_category;
```

Output Preview:

user_category	num_users	total_posts	avg_engagement_per_post
High Follower (>= 25k)	742	6106	3787.75
Low Follower (< 25k)	758	6254	3768.59

3. Question H6: Find the Most Suspicious High Impact Users

Logic: Deployed Common Table Expressions (CTEs). The first CTE isolated global average engagement. The second aggregated user-level metrics and flagged suspicious posts (shares > likes). The final query cross-joined these tables to isolate low-follower users (<10,000) exhibiting artificially inflated metrics.

SQL Query:

```
WITH OverallAvg AS (SELECT AVG(likes + shares + comments) AS global_avg FROM social_data),
UserStats AS (
    SELECT user_id, location, follower_count, COUNT(post_id) AS num_posts,
    AVG(likes + shares + comments) AS avg_engagement, SUM(likes + shares + comments) AS total_engagement,
    SUM(CASE WHEN shares > likes THEN 1 ELSE 0 END) AS suspicious_posts
    FROM social_data GROUP BY user_id, location, follower_count
)
SELECT u.user_id, u.location, u.follower_count, u.num_posts, u.avg_engagement, u.total_engagement
FROM UserStats u CROSS JOIN OverallAvg o
WHERE u.follower_count < 10000 AND u.avg_engagement > o.global_avg AND u.suspicious_posts > 0
ORDER BY u.total_engagement DESC LIMIT 3;
```

Output Preview:

user_id	location	follower_count	num_posts	avg_engagement	total_engagement
user_cssz1rea	Tokyo, Japan	7863	16	4048.50	64776.0
user_cuzig14g	Cairo, Egypt	8070	16	4002.50	64040.0
user_hdas0iau	Rio de Janeiro, Brazil	1620	16	3990.43	63847.0